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ORIGINAL ARTICLE

Applying a meta-heuristic algorithm to predict and optimize compressive strength of concrete samples

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Abstract
The successful use of fly ash (FA) and silica fume (SF) materials has been reported in the design of concrete samples in the literature. Due to the benefits of using these materials, they can be utilized in many industrial applications. However, the proper use of them in the right mixes is one of the important factors with respect to the strength and weight of concrete. Therefore, this paper develops relationships based on meta-heuristic (MH) algorithms (artificial bee colony technique) to evaluate the compressive strength of concrete specimens using laboratory experiments. A database comprising silica fume replacement ratio, fly ash replacement ratio, total cementitious material, water content coarse aggregate, high-rate water-reducing agent, fine aggregate, and age of samples, as model inputs, was used to evaluate and predict the compressive strength of concrete samples. Developed models of the MH technique created relationships between the mentioned parameters. In the new models, the influence of each parameter on the compressive strength was determined. Finally, using the developed model, optimum conditions for compressive strength of concrete samples were presented. This paper demonstrated that the MH algorithms are able to develop relationships that can serve as good substitutes for empirical models.

Keywords Compressive strength of concrete · Meta-heuristic algorithms · Artificial bee colony · Optimization technique

1 Introduction

Substitute materials such as fly ash (FA) and silica fume (SF) have been introduced as cheap cement material than the original cement in concrete [1]. These two materials have been used in concrete, due to recent research on their properties [2]. Due to the fact that various materials are produced from other industrial wastes, the optimal use of some of them leads to lower costs and more environmental protection.

Using FA, about 20–50% of the total volume of cementitious aggregates as part of cement in concrete samples has become common in construction works [3]. In addition, the initial strength factor is an effective feature in concrete, which, if it is not the first priority, can increase the amount of FA by up to 60%. This is because of the negative impact that FAs have on concrete strength and are considered in different work. However, if lower water to binder ratios could be used, the strength of these concrete against loading conditions could be improved [4].

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SF materials can be used as filler and pozzolan in concrete samples due to the presence of fine particles [5]. According to study conducted by Mazloom et al. [6], by increasing the percentage of SF, short-term resistance, such as 28-day resistance, improves on concrete specimens, but this results in problems with the workability of concrete in comparison with lower values. However, the percentage of SF in concrete samples that can be obtained from optimum performance is still being investigated by other researchers [4–6]. Several researchers have provided different limits for using SF to obtain the optimum concrete strength, experimentally [7, 8]. In addition, some other works have been done to obtain compressive strength of concrete [6, 7]. These include cementitious parameters such as FA and SF. According to old perspective, effective parameters were assumed to be based on analytical relationship between the compressive strength of concrete and other parameters in laboratory samples and evaluated the unknown values using statistical solutions such as regression analysis [8]. One of the problems with these methods is that they are not flexible in different specimens, which makes it difficult to estimate the compressive strength of the concrete with precision and in all models. In addition, the effect of each of the parameters on the compressive strength of the concrete is not easily carried out. In recent years, modeling of various phenomena has been accomplished using meta-heuristic algorithms and artificial intelligence systems such as artificial neural network (ANN) that were used in various engineering disciplines [9–44]. ANNs are new computing programs that are capable of solving linear and nonlinear problems with different degrees of indeterminate. Problems are processed using neurons that are similar to the human brain, which have several computational layers [15, 45–47]. In these systems, predictive models are developed for the behavior of materials based on laboratory data. One of the important points in the dataset is that the data should contain parameters that affect the desired behavior. In this condition, the ANN can obtain the desired behavior with the selected parameters. With these new models, in addition to predicting, the results of other experiments can be done using generalizability. This new approach in civil engineering has been developed and expanded in other fields as well (e.g., [48–56]).

The purpose of this paper is to combine two models for the prediction and optimization of the compressive strength of concrete, taking into account the long-term behavior of FA and SF parameters. Therefore, the implementation of a new algorithm for model presentation and then optimization is discussed in this paper. The required data were gathered from 144 actual tests based on concrete mix design. This set has eight parameters that affect the compressive strength of the concrete including SF replacement ratio, FA replacement ratio, total cementitious material (TCM), water content (W) coarse aggregate (CA), high-rate water-reducing agent (HRWRA), fine aggregate (SSA), and age of samples (AS). Prediction and optimization models are developed in different parts of the model. In the following parts, after giving descriptions of database, principles of the used intelligent systems will be explained. Then sections of model development with details will describe for prediction and optimization of the compressive strength of concrete and eventually two new models will be introduced in the field of concrete technology.

2 Database

A set of 144 laboratory data was collected from various investigations to evaluate the compressive strength of the concrete for this study. The set was selected on the basis of eight effective parameters on concrete strength. In all experiments, samples were selected that had similar test conditions and parameters. One of the most important parts of intelligent models is the preparation of input and output data. Therefore, the data used after the initial analysis were selected based on their influence on compressive strength of the concrete. Input parameters of smart models include SF, FA, TCM, W, CA, HRWRA, SSA, and AS. Figures 1, 2, 3, 4, 5, 6, 7 and 8 show the distributions of the input parameters for all 144 samples.

This research has been developed to overcome the limitations of the experimental relationships used for the compressive strength of concrete samples. The limitations of

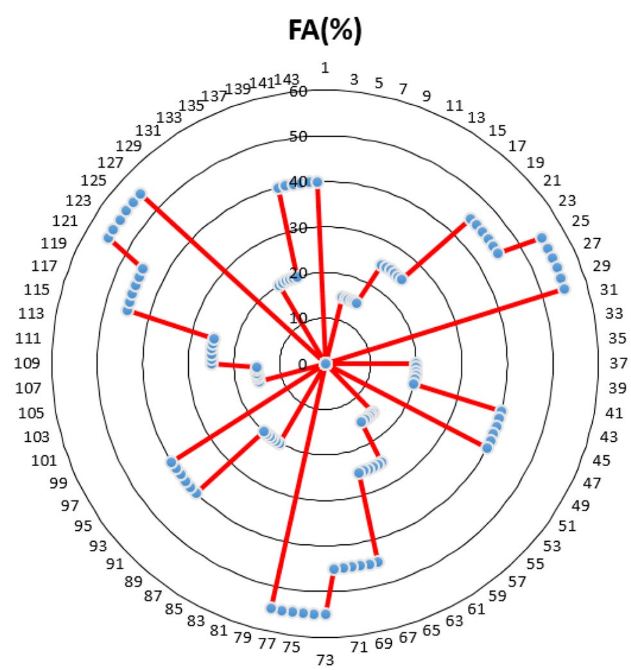

Fig. 1 The FA parameter distribution for all 144 laboratory samples

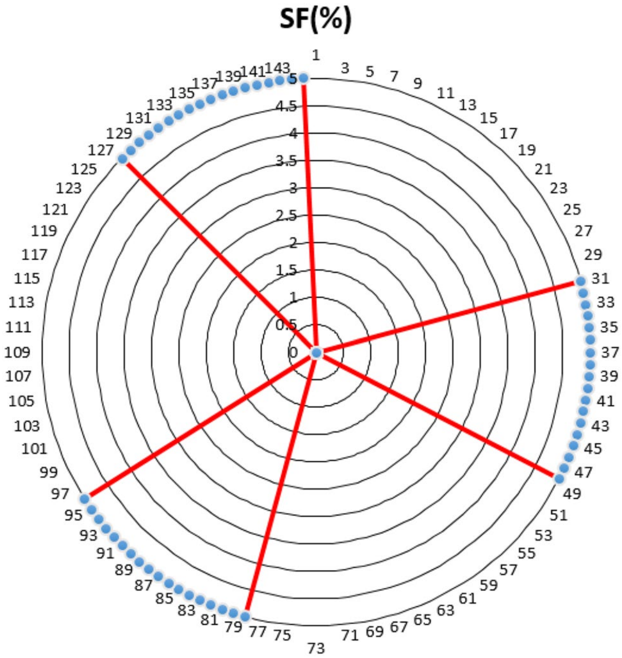

Fig. 2 The SF parameter distribution for all 144 laboratory samples

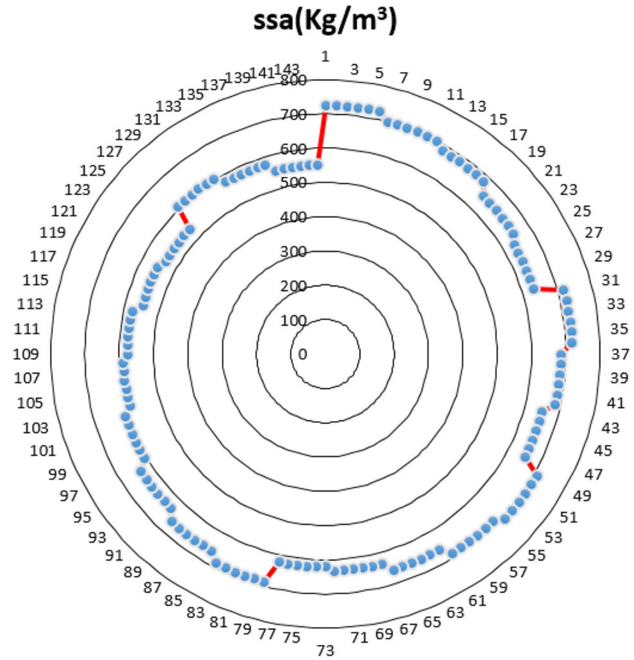

Fig. 4 The ssa parameter distribution for all 144 laboratory samples

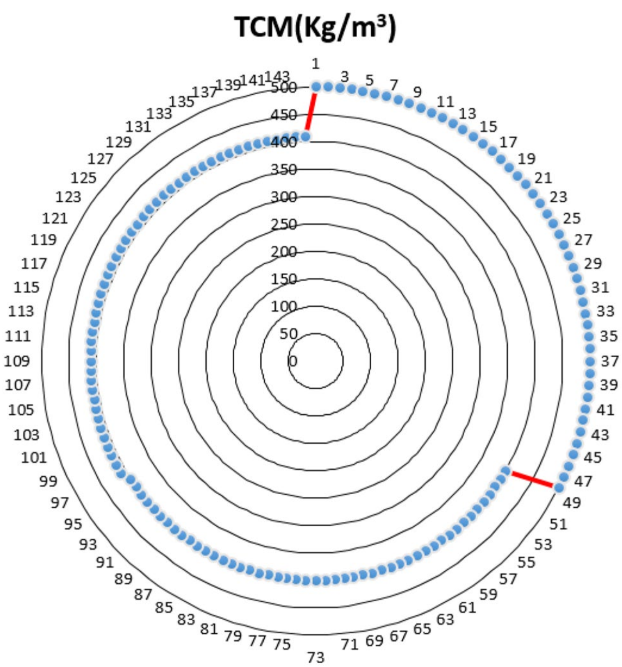

Fig. 3 The TCM parameter distribution for all 144 laboratory samples

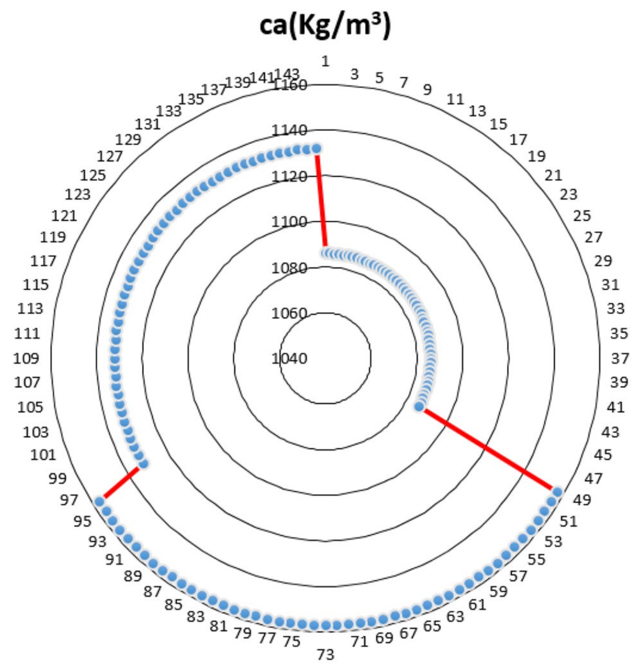

Fig. 5 The ca parameter distribution for all 144 laboratory samples

experimental relationships are limited to specific laboratory conditions, relatively low accuracy models, disregarding various parameters, and low reliability in engineering design and optimization. Therefore, this research implements intelligent models as a new solution in this field. It takes into account the impact of different laboratory parameters and ultimately provides a new capability for designers to implement different conditions of the optimized designs.

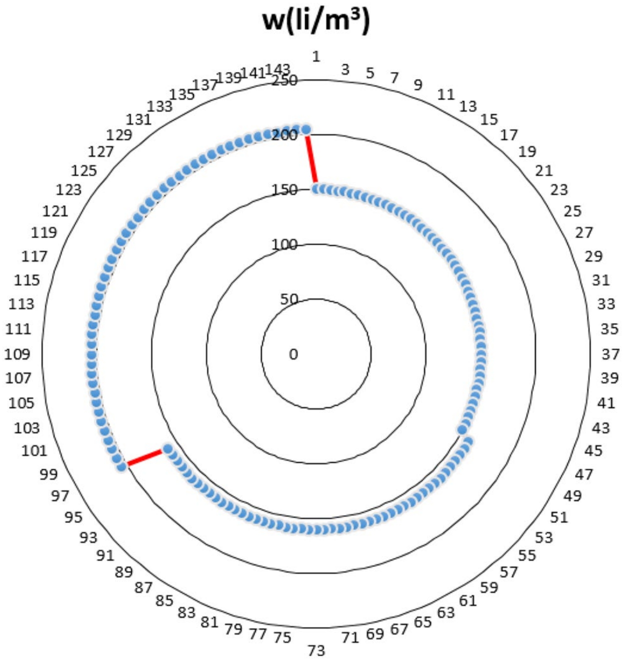

Fig. 6 The w parameter distribution for all 144 laboratory samples

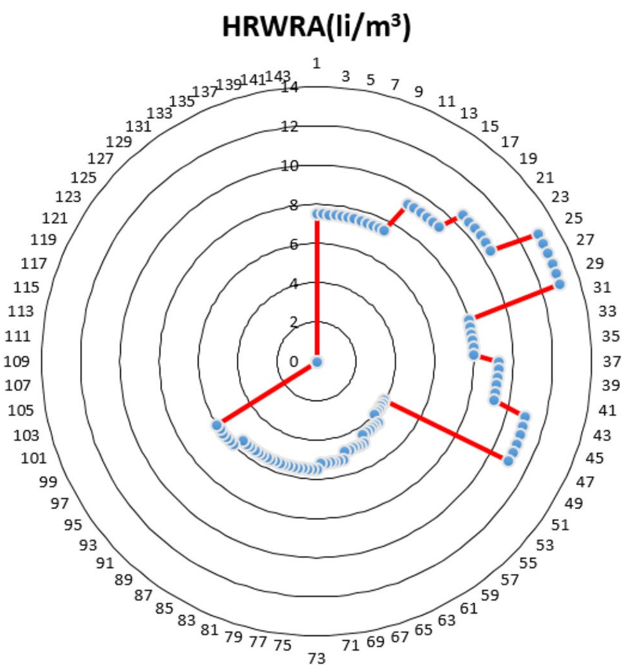

Fig. 7 The HRWRA parameter distribution for all 144 laboratory samples

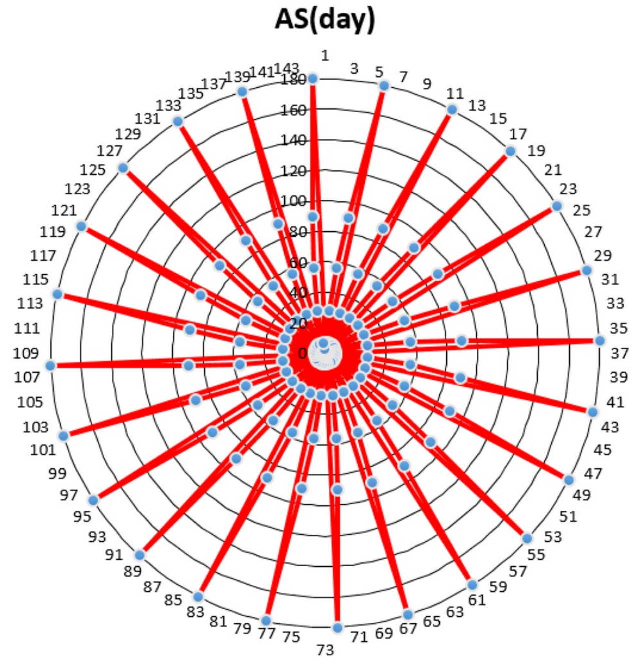

Fig. 8 The AS parameter distribution for all 144 laboratory samples

3 Research methodology

3.1 Artificial neural network

The neural network consists of functional set containing neural structures along with the central system. Actually, this network follows the human brain's nerve specification, for example, uses the power of learning and generalization of previous examples for later work [17, 57]. At the beginning of research, Frank Rosenblatt devised a machine that was similar to the human mind and became known as Perceptron. This machine has a continuous memory, consisting of several sections, including the input, output and hidden layers. All layers are interconnected, each of which is assigned a weight to get the desired output at the end [27, 28, 58–60]. The center of this section, known as the "training" of the neural network, is learning these algorithms for proper weighting. Perceptrons have been used as a quick and simple solution in various complex neural networks [61]. Figure 9 shows the overall schema of this network. In the following, the weights of the networks are obtained using the following equation:

$$\mathrm{Net}_j = \sum_{i=1}^{n} w_{ij} x_{ij}. \tag{1}$$

In this relation, Net_j is introduced as the weight of the jth nerve for the input layer obtained from the previous

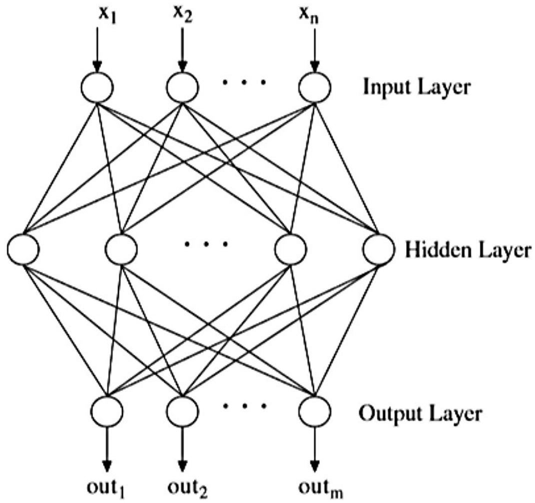

Fig. 9 The structure of a multi-layer perceptron grid for data analysis

layer, which contains the n neuron. In this equation, w_{ij} is the weight between the jth neurons and the ith neurons in the previous layer and x_i is the output of the ith neuron in the previous layer. Finally, the output of the jth neuron, shown with out_j, is obtained from the following equation:

$$\text{out}_j = f(\text{Net}_j) = \frac{1}{1 + \exp(-k\text{Net}_j)}, \quad (2)$$

where k is for controlling semi-linear gradient and has a constant value. The sigmoid activator function can be used in other layers except the input layer that is related to the data [62]. One of the features of neural networks is the fast processing of data in any situation. Additionally, these networks are able to provide different degrees of output, even if the data are incomplete or has error [63, 64]. The back-propagation network is one of the most commonly used structures in the field of artificial intelligence, which is used in classification, forecasting and planning processes. This structure, which is required for learning networks, actually evaluates and then reduces model errors by sending a stream from the input layer to the external one [65]. The neural network performs five steps in designing models: creating data sets, designing a first structure, estimating training processes, training networks and examining their performance for model adaptability [66]. Figure 10 shows a developed structure of the current research.

3.2 Artificial bee colony

The bee colony algorithm, which is proposed to optimize the real parameters, is the optimization algorithm. It follows the collective behavior of the bee colony and it is simulated accordingly [67, 68]. The bee colony algorithm categorizes

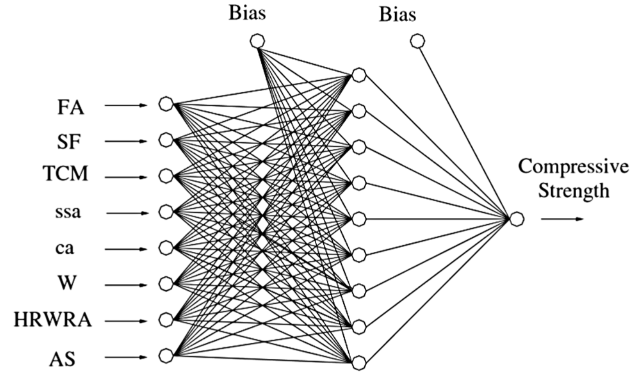

Fig. 10 ANN structure of the developed model

the bee swarm into three categories, that is, the employed bees, onlooker bees and scout bees. Half of the colony is comprised of employed bees and the other half includes onlooker bees. The employed bees forage for food source in their mind, while transferring their food information to onlooker bees. The onlooker bees tend to find the desirable food source from among what the employed bees have found. The scout bees are defined by some employed bees that have left their food sources and forage new sources.

Like other population-based algorithms, the bee colony algorithm is an iterative process [69]. The primary units of bee colony algorithm can be explained as follows:

1. Population initialization and evaluation.
2. Employed bee phase.
3. Probabilistic selection task.
4. Onlooker bee phase.
5. Scout bee phase.
6. In case the termination condition is not realized, go back to phase 2; otherwise, end.

3.2.1 Population initialization phase

The initial population of the solution is filled with SN real-valued vectors with n dimensions that are randomly selected (i.e., food sources):

$$X_i = \{x_{i1}, x_{i2}, \ldots, x_{in}\}, \quad (3)$$

X_i represents the ith food source in the population. The other food sources are produced as follows:

$$x_{ij} = x_{\min j} + \text{rand}(0, 1)\left(x_{\max j} - x_{\min j}\right), \quad (4)$$

where $i = 1, 2, \ldots, \text{SN}$, $j = 1, 2, \ldots, n$, $x_{\min j}$ and $x_{\max j}$ represent the lower and upper thresholds of j dimension, respectively. These food sources are randomly dedicated to SN-employed bees and their proportion is evaluated.

3.2.2 Employed bee phase

At this phase, each X_i employed bee produces a v_i new food source near its current location using the search equation provided below:

$$v_{ij} = x_{ij} + \emptyset_{ij}\left(x_{ij} - x_{kj}\right), \tag{5}$$

where $k \in \{1, 2, \dots, \mathrm{SN}\}$ and $J \in \{1, 2, \dots, n\}$ are randomly selected indices; K should be different from i; $\emptyset_{ij}$ is a random real number normally distributed in $[-1, 1]$.

As v_i is achieved, it is evaluated and compared with X_i. If v_i is equal or better than X_i, then v_i replaces X_i and becomes a new member of the population; otherwise, X_i is kept. In other words, the greedy selection mechanism is used between the old and candidate solutions.

3.2.3 Estimation of the probable values involved in probabilistic selection

After finishing the search task, the employed bees share their information about the amount of nectar and the location of the food source with onlooker bees in the field. The onlooker bee evaluates the information received from all the employed bees and selects a food source with the probability related to the amount of nectar.

This selection is based on the proportionality data solution in population. The proportionality-based selection can be the roulette wheel, ranking by random sampling, championship selection and/or other selection designs. In initial ABC, the roulette wheel selection, in which each section is proportionate to the corresponding value, is used as follows:

$$P_i = f_i / \sum_{j=1}^{\mathrm{SN}} f_j, \tag{6}$$

where f_i is the proportionality value of solution i. Clearly, the greater the value of f_i is, the greater the selection probability of ith food source is.

In fact, the amount of nectar in the food source shows the fitness of the problem response.

In ABC algorithm, the fitness function is defined as follows:

$$F_i = \begin{cases} \frac{1}{1+F(x_i)} & F\left(x_i\right) \geq 0 \\ 1 + \left|F\left(x_i\right)\right| & F\left(x_i\right) < 0 \end{cases}, \tag{7}$$

where $F(x_i)$ is the value of the target function of ith response and f_i is the fitness of ith response after producing new responses.

3.2.4 Onlooker bee phase

The onlooker bee evaluates the information received from the employed bees regarding the amount of nectar and selects X_i food source with regard to the probable value of p_i. As soon as the onlooker bee selects X_i food source, it applies corrections on X_i by the use of Eq. (5). Like the employed bees, if the corrected food source has a nectar value equal or better than X_i, the corrected food source replaces X_i and becomes a new member of the population.

3.2.5 Scout bee phase

If X_i food source cannot be improved over a preset level, the food source will be discarded and the employed bees become scout bees, such that they act as scouts with regard to the discarded source. The scout bees will produce the food source randomly in the problem limits as follows:

$$X_{ij} = x_{\min j} + \text{ rand } (0, 1)\left(x_{\max j} - x_{\min j}\right), \tag{8}$$

where $j = 1, 2, \dots, n$.

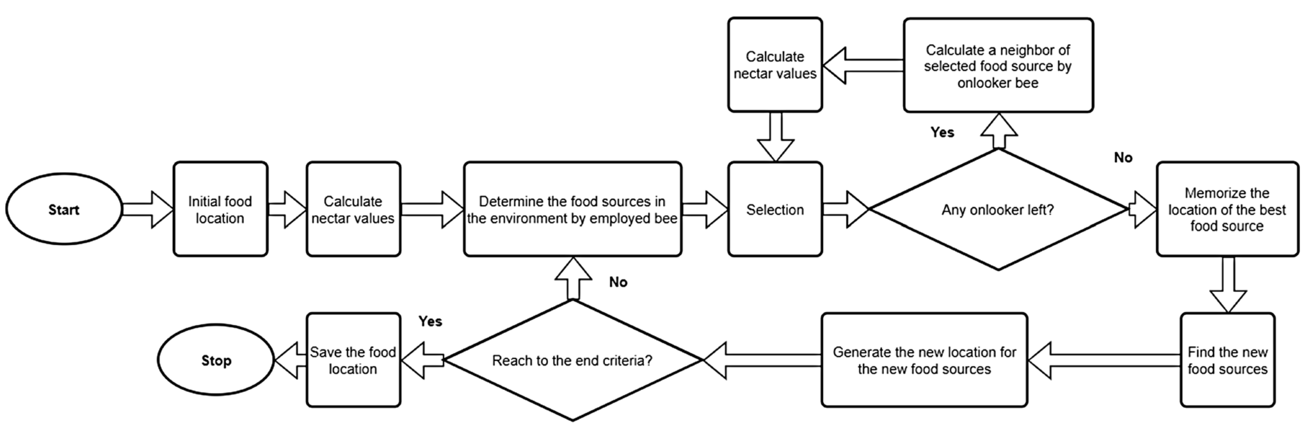

Fig. 11 Total process of ABC algorithm [48]

Figure 11 displays a flowchart of the general processes for the new ABC algorithm.

4 Modeling

4.1 ANN modeling

Development of artificial intelligence-based models can provide more knowledge of the contents of laboratory data. Due to the importance of concrete in civil engineering, the use of new techniques has led to the recognition of its various parameters and the development of patterns that can make designs more accurate. New computational techniques can improve the accuracy of models and designs and provide more advanced capabilities. In this research, inspired by these new needs, intelligent design was implemented and finally the solutions used with optimization techniques.

In this research, input and output data were selected according to the data presented in Sect. 2. Various models of ANN were designed for prediction purposes. To design the overall neural network, based on the suggestions of previous researchers, three layers were considered as the base of simulations [70]. Then the effect of two main parameters including the number of replicates and the number of neurons in the second layer (hidden layer) was evaluated on the performance of the results. The number of iterations in this study was between 50 and 400. The number of neurons, which is one of the most important parts of the network, was chosen between 1 and 15. These ranges were based on the recommendations of previous investigations [71–73]. For designing neural networks, 80% of all available data was used to train the system while the remaining 20% was utilized to test/evaluate the system. Statistical indices were used to compare the performance of the systems. Here are two important relationships that are used to evaluate ANN models [74, 75]:

$$R^2 = 1 - \frac{\sum_{i=1}^{N} (O_i - T_i)^2}{\sum_{i=1}^{N} (T_i - \overline{T}_i)^2}, \tag{9}$$

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{N} (O_i - T_i)^2}{N}}, \tag{10}$$

where T_i and O_i are real and predicted values, respectively, which represent mean values of T_i. N is the number of total data. Finally, the results of different ANN models were obtained. The results of five sets of data for the prediction of compressive strength of the concrete are given in Table 1. To select the best ANN model, based on the average values of coefficient of determination (R^2) for training and testing sections, Fig. 12 should be considered. As it can be seen, model no. 5 with five neurons shows the best performance in predicting compressive strength of the concrete.

Table 1 Results of the ANN models for the prediction of compressive strength of the concrete

Model	No. of hidden nodes	Set 1		Set 2		Set 3		Set 4		Set 5		Average	
		R^2		R^2		R^2		R^2		R^2		R^2	
		Train	Test	Train	Test	Train	Test	Train	Test	Train	Test	Train	Test
1	1	0.849	0.756	0.920	0.770	0.942	0.891	0.943	0.894	0.943	0.894	0.919	0.841
2	2	0.978	0.967	0.900	0.844	0.951	0.926	0.857	0.544	0.981	0.969	0.934	0.850
3	3	0.390	0.125	0.988	0.972	0.974	0.934	0.991	0.956	0.985	0.948	0.865	0.787
4	4	0.969	0.934	0.784	0.750	0.994	0.971	0.990	0.971	0.996	0.993	0.946	0.924
5	5	0.990	0.969	0.991	0.944	0.991	0.904	0.993	0.888	0.995	0.972	0.992	0.935
6	6	0.395	0.125	0.993	0.808	0.996	0.968	0.996	0.971	0.998	0.942	0.875	0.763
7	7	0.997	0.990	0.993	0.970	0.996	0.743	0.999	0.964	0.998	0.805	0.997	0.894
8	8	0.994	0.969	0.830	0.618	0.997	0.967	0.999	0.867	0.999	0.927	0.964	0.870
9	9	0.100	0.103	0.993	0.937	0.998	0.955	0.998	0.889	0.998	0.792	0.817	0.735
10	10	0.998	0.979	0.998	0.816	0.995	0.791	0.999	0.911	1.000	0.931	0.998	0.886
11	11	0.719	0.489	0.998	0.973	0.998	0.861	0.999	0.950	1.000	0.804	0.943	0.815
12	12	0.990	0.928	0.999	0.814	0.999	0.936	1.000	0.780	1.000	0.927	0.998	0.877
13	13	0.702	0.683	0.999	0.942	0.998	0.531	1.000	0.877	1.000	0.953	0.940	0.797
14	14	0.997	0.961	1.000	0.814	0.840	0.579	1.000	0.945	0.590	0.466	0.885	0.753
15	15	0.460	0.173	0.999	0.956	1.000	0.953	1.000	0.940	1.000	0.276	0.892	0.659

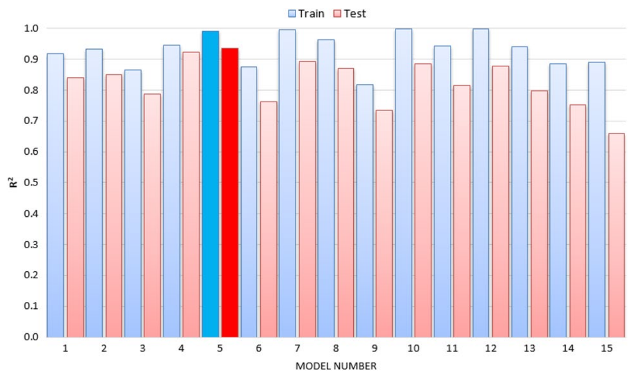

Fig. 12 The result of average R^2 for different ANN models with various number of hidden nodes

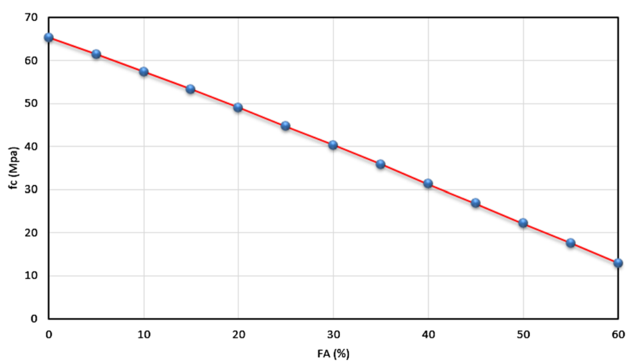

Fig. 13 Results of FA changes on f_c based on the newly developed model

4.2 Evaluating the compressive strength of concrete

As mentioned before, model no. 5 of ANN performs better than other models to predict compressive strength of concrete. Considering this model, the behavior of different parameters on the compressive strength of concrete is investigated. To evaluate the effect of each parameter, a fixed laboratory sample is assumed and the parameter changes in the certain intervals.

Figures 13, 14, 15, 16, 17, 18, 19 and 20 show the results of the behavior of various parameters on the compressive strength of concrete based on the developed model. It is also assumed that the values of the other parameters in each figure are constant. As can be seen in Fig. 13, if the percentage of FA increases, the amount of compressive strength of concrete decreases linearly. The same changes also occur due to the increased SF percentage in the samples (Fig. 14). These two parameters indicate that they cause a significant reduction in the compressive strength of concrete. Therefore, it is recommended to use low percentages of both materials in structures requiring high strength.

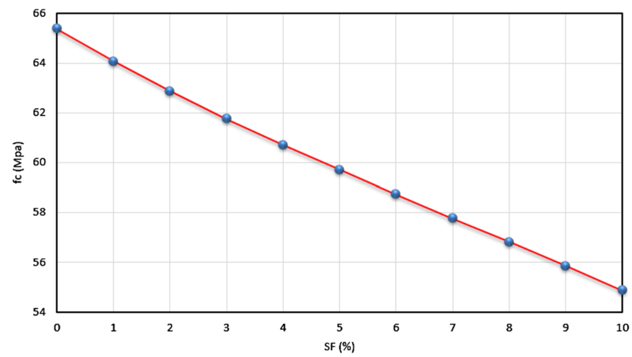

Fig. 14 Results of SF changes on f_c based on the newly developed model

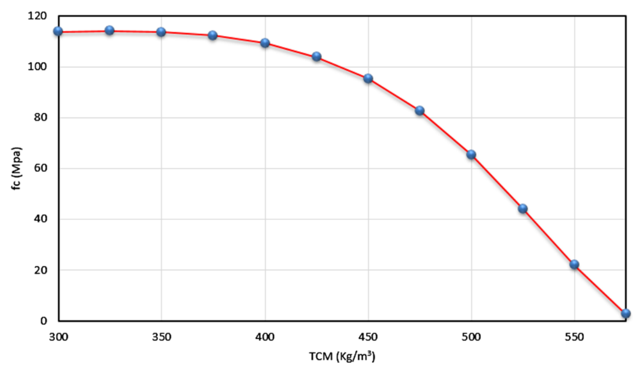

Fig. 15 Results of TCM changes on *fc* based on the newly developed model

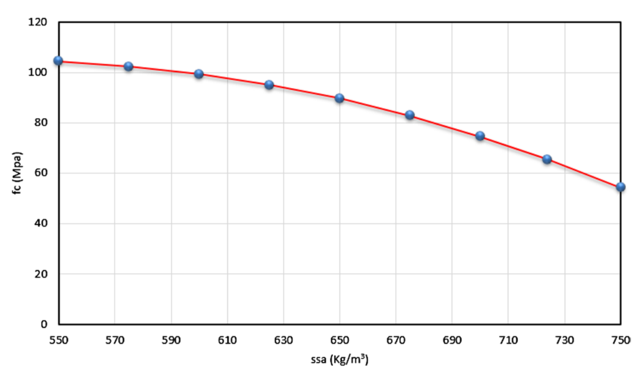

Fig. 16 Results of ssa changes on f_c based on the newly developed model

The TCM parameter behaves differently depending on the amount consumed. Up to values below 400, the samples are highly resistant but with increasing TCM, the trend decreases. If this value is increased, the compressive strength of concrete also decreases more rapidly (Fig. 15). The parameters ssa, ca and w also decrease by increasing value with a gentle slope and become steeper at high values (Figs. 16, 17 and 18). By examining Fig. 19, it can be concluded that the trend

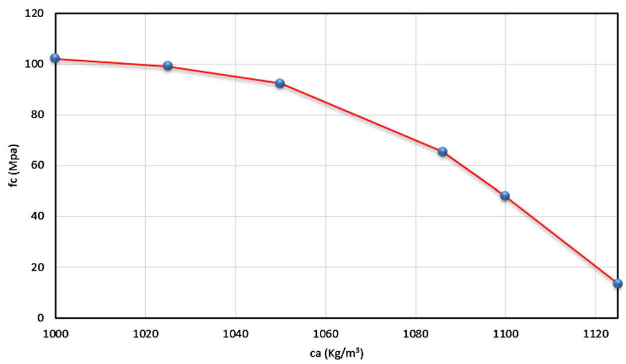

Fig. 17 Results of ca changes on f_c based on the newly developed model

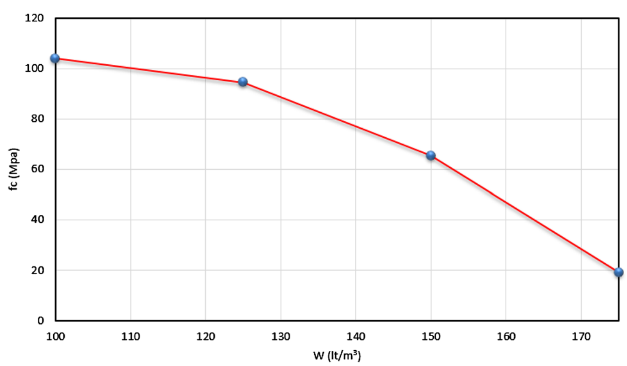

Fig. 18 Results of W changes on f_c based on the newly developed model

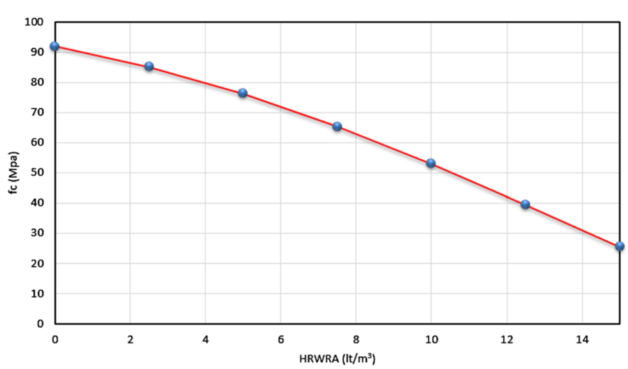

Fig. 19 Results of HRWRA changes on f_c based on the newly developed model

of HRWRA changes on compressive strength of concrete is decreasing. The variation of this parameter on the sample strength is high. Therefore, it can be more sensitive. Finally, increasing trends can be observed by examining the trend of age changes in sample strength (Fig. 20). This trend indicates that resistance increases at a higher rate in the early days and after 90 days, and when resistance reaches a high amount, the

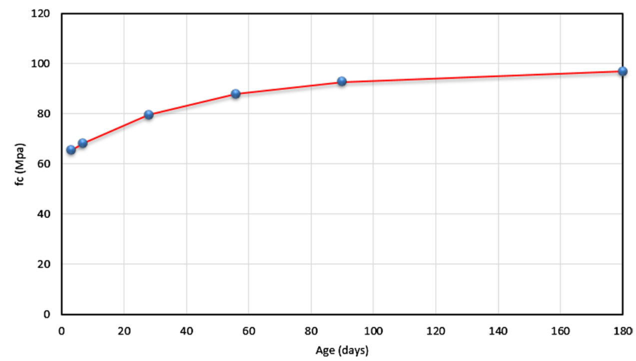

Fig. 20 Results of age changes on f_c based on the newly developed model

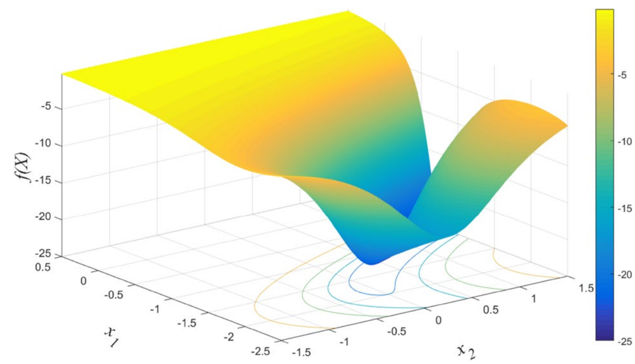

Fig. 21 The three-dimensional graph of Eq. 11

changes decrease. This represents the correct trend of models developed by artificial intelligence.

5 Optimization modeling

To examine ABC algorithm, which is used in this research for optimizing compressive strength of concrete samples, the selected functions were employed. Here, two functions were presented according to Eqs. 11 and 12 as follows. The minimum values of these functions in the mentioned intervals are − 25 and − 0.3523, respectively. Figures 21 and 22 illustrate the three-dimensional graph of these two functions in the specific interval. Figures 23 and 24 demonstrate the results obtained through ABC algorithm, which is for these two equations. As can be seen, the written code of this algorithm can identify the minimums well. That is why this code can be run for this research and the results obtained in the previous section.

$$F_1(x) = \frac{-100}{10\left[\left(x_1+1\right)^2 - \left(x_2+1\right)^2\right] + x_1^2 + 4}, \tag{11}$$

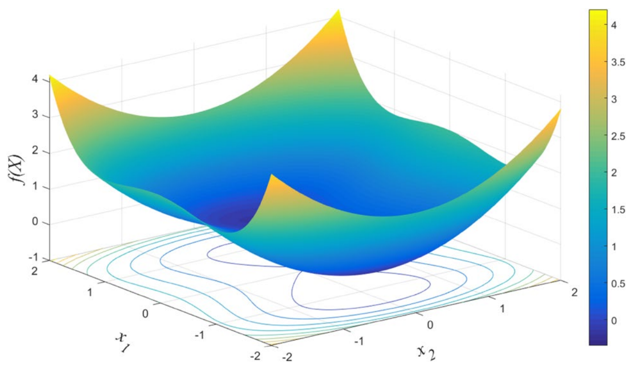

Fig. 22 The three-dimensional graph of Eq. 12

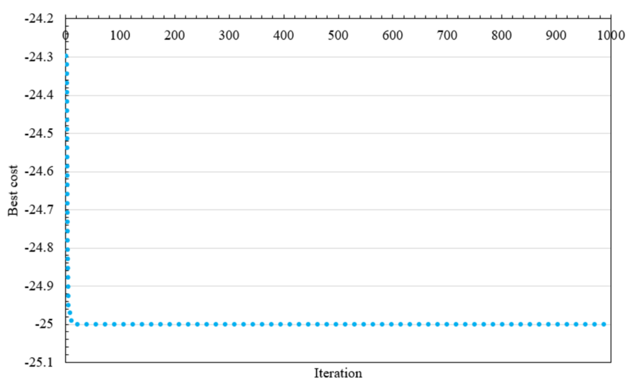

Fig. 23 The result of ABC algorithm for Eq. 11

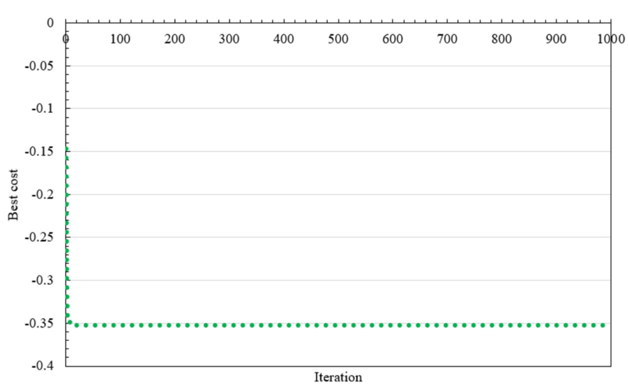

Fig. 24 The result of ABC algorithm for Eq. 12

$$F_2(x) = 0.25x_1^4 - 0.5x_1^2 + 0.1x_1 + 0.5x_2^2 \sin(x_1 + x_2) + (x_1 + x_2)^2 - 1.5x_1 + 2.5x_2 + 1. \tag{12}$$

After first examining the performance of the ABC algorithm in solving basic equations, we implemented and combined this algorithm with the selected ANN model (model no. 5) that is introduced as the best predictor for the compressive strength of concrete. The optimization algorithm was implemented to determine the optimal parameters, introduce different design conditions and finally design the best concrete pattern parameters. Since the ABC algorithm needs to adjust the initial conditions, an initial analysis was performed to obtain the best conditions. These analyses were performed by trial and error to obtain the best initial conditions. Finally, 35 bees were selected as suitable populations. This process was also performed to obtain the iteration parameter and 350 iterations were selected as the optimal response mode.

In this problem, eight parameters were used to design a forecasting model. Since all parameters can be changed without regard to actual conditions, problems with different properties can be designed. Therefore, to illustrate the design of optimal conditions in this research, several cases were evaluated based on the actual conditions of the parameters. One of these conditions is to investigate the effect of the age parameter on the amount of compressive strength of concrete. For these conditions, 24 samples were examined for two 90- and 180-day conditions. Different examples for these two times offer different resistances depending on the material conditions.

Using the developed model and the optimization algorithm, all samples were analyzed for these two times and the optimum case was calculated which is the maximum value that can be obtained for the compressive strength of concrete. Table 2 presents the optimal results along with 24 laboratory samples for both 90 and 180 times.

In general, the resistance values for the 180-day samples are greater than 90 days. In addition, the optimized examples obtained by the optimization algorithm provide high performance. As can be seen, the maximum values of the samples obtained by optimization are higher than the maximum of the experiments. The table also shows that the mean averages of samples optimized for 90 and 180 times were 103.5761 and 112.1919, respectively, which were significantly better than those of laboratory samples (70.525 and 75.525, respectively). Therefore, by optimizing the laboratory specimens, a reasonable improvement in the performance of the specimens can be achieved.

6 Conclusions

Two important goals were taken in this paper. The first goal is to develop neural models for predicting the compressive strength of concrete. The second goal is to combine the prediction model and optimization algorithm for optimal purposes, and especially design different conditions to improve the performance of concrete shear strength. Given the importance of input data to the system, a dataset was collected from laboratory samples. Important and influential parameters of the applied models were identified and their

Table 2 Experimental and optimized values of samples according to age parameter

Sample	90 days		180 days	
	Experimental	Optimized	Experimental	Optimized
1	95.7	93.5257	97.7	112.0638
2	99.6	110.2095	106.3	119.7245
3	95.4	104.449	107.8	125.8991
4	87.7	99.7563	97.7	120.1954
5	72.8	108.5805	79.9	114.5499
6	93.6	102.1826	99.3	101.102
7	90.3	97.5535	95.9	112.9365
8	83.4	107.894	88.3	113.6321
9	70.5	102.5645	70.6	107.9045
10	68.5	101.5689	72.1	109.8107
11	66.2	104.3992	70.2	111.1084
12	61.2	109.5865	63.7	107.2127
13	52.9	98.3678	63.2	114.3768
14	73.7	101.9895	74.5	99.3719
15	72.4	113.5166	76	118.0455
16	59.1	98.781	68.4	112.452
17	58.1	110.1692	60.6	102.7638
18	62.6	111.3825	64.8	119.8144
19	53.7	96.8418	57.9	107.1186
20	54.1	98.4426	56.6	113.6102
21	41.4	102.0645	48.4	120.7809
22	67.3	106.9001	66.3	111.6498
23	63.7	99.4851	68	103.887
24	48.7	105.6164	58.4	112.5953
Min	41.4	93.5257	48.4	99.3719
Max	99.6	113.5166	107.8	125.8991
Ave	70.525	103.5761	75.525	112.1919
STD	16.0773	5.1878	16.7337	6.4913

effects were discussed with ANN and then ABC algorithm. The results obtained from this study are presented below:

- Adding SF and FA causes to decrease initial resistance as well as the final strength of the concrete; therefore, they should be used in low percentages and in accordance with the need for resistance.
- The behavior of TCN value is different in concrete samples. High values of TCN can cause reduction in compressive strength of concrete.
- ANN model was implemented for different conditions. Finally, the final model could establish a suitable relationship between concrete compressive strength and effective parameters with high precision.
- The ANN model was combined with the ABC optimization algorithm and optimized conditions to increase the compressive strength of the concrete. Using this system, optimal parameters can be achieved in design with better accuracy and speed.

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